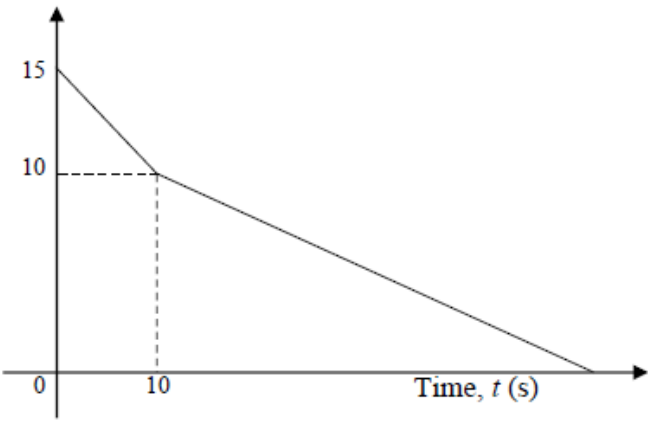
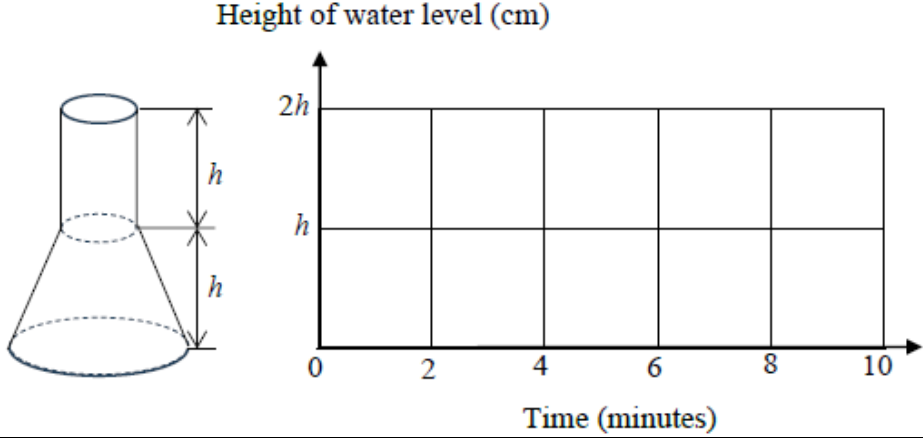
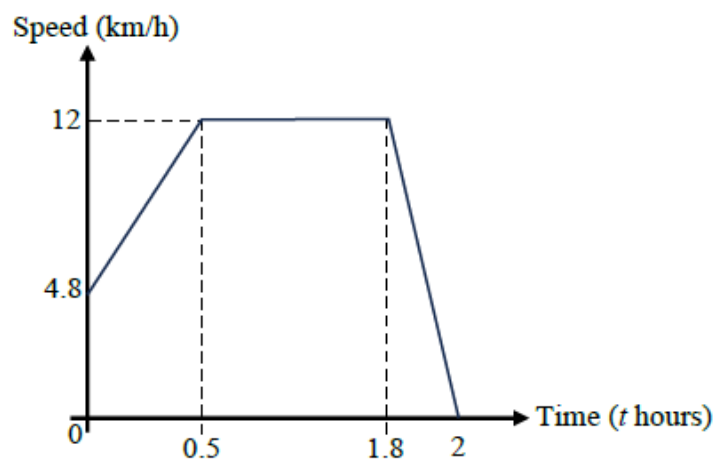
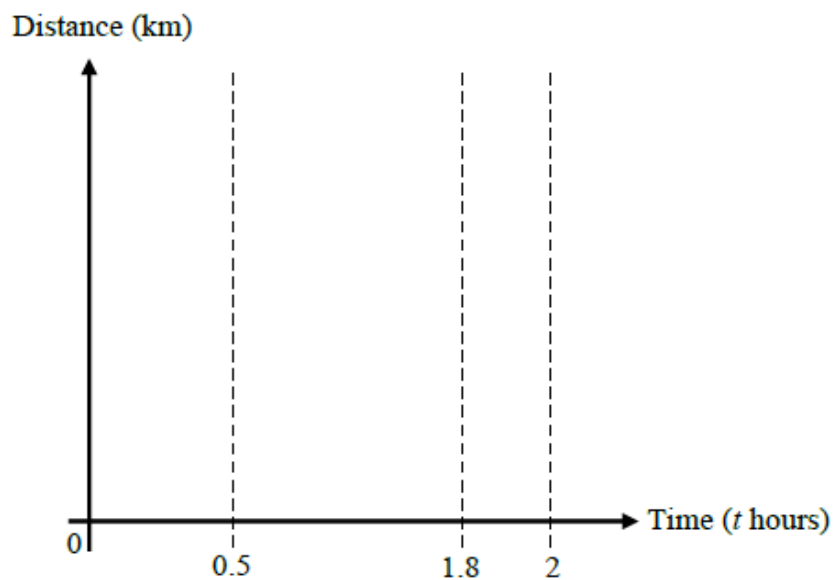


1	YTSS/2025/I/18
	The diagram shows the speed-time graph of a train as it approaches the station. It decelerates uniformly from 15 m/s to 10 m/s in 10 seconds. Speed (m/s)  (a) Calculate the deceleration during the first 10 seconds. [1] (b) Find the speed of train at $t = 4$. [1] (c) At $t = 10$, the train driver notices that there is a foreign object on the track about 100 metres away. If the train decelerates uniformly at 0.4 m/s^2 until it comes to a stop, determine if he is able to stop without hitting the object. Explain with working. [2]
2	SCGS/2025/I/17
	The flask in the diagram is initially empty. It is filled with water flowing at a constant rate from a tap. The water level takes 7 minutes to reach a height of h cm and 3 more minutes to fill the flask completely. On the grid, sketch the graph of the height of water level against time. Height of water level (cm)  Time (minutes) [2]

The diagram shows the speed-time graph of a runner 2 hours after crossing a checkpoint.

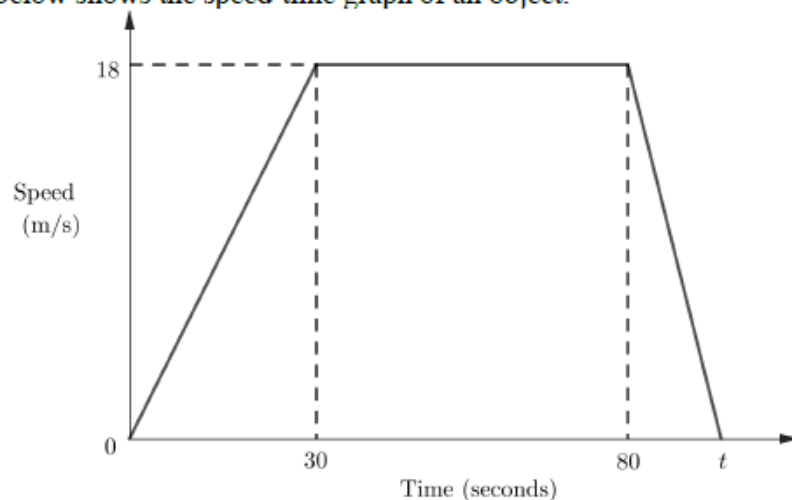


- (a) Calculate the speed of the runner at $t = 0.2$. [2]
- (b) Calculate the average speed of the runner during the 2 hours. [3]
- (c) On the axes below, sketch the distance-time graph of the runner after crossing the checkpoint.



[3]

The diagram below shows the speed-time graph of an object.



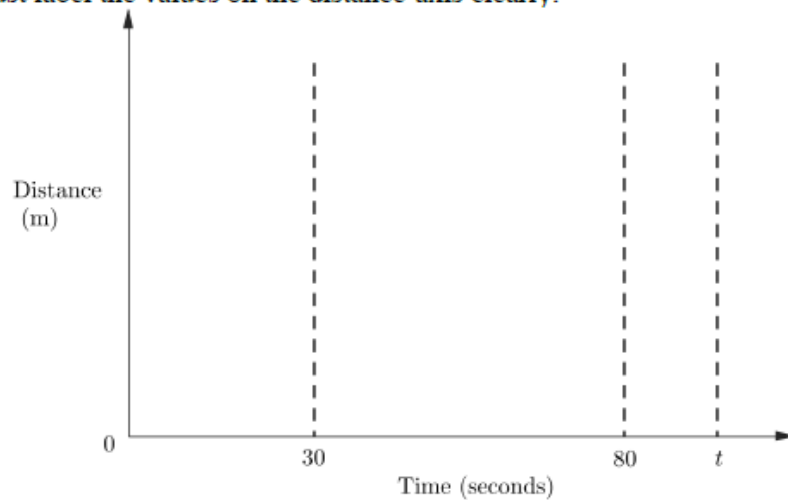
(a) Calculate the speed of the object at 18 seconds. Give your answer in km/h.

[2]

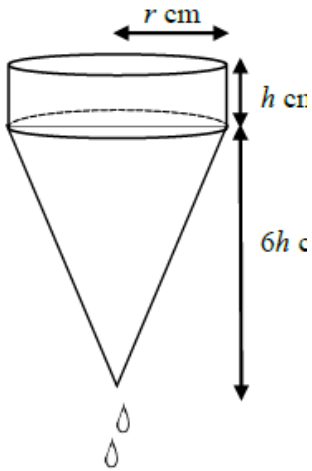
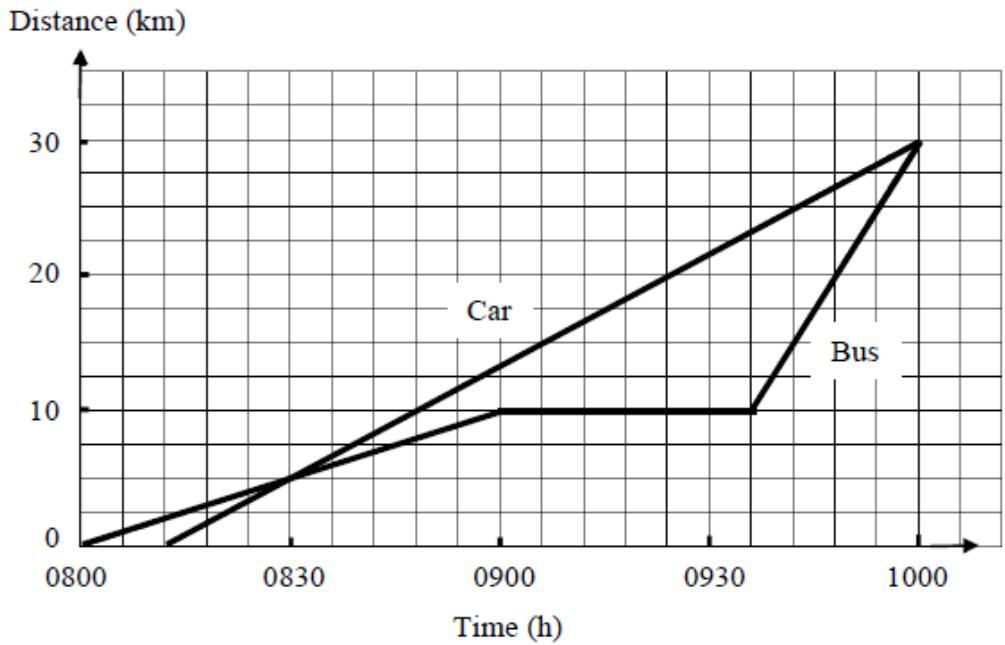
(b) At 80 seconds, the object starts decelerating at a rate that was twice its acceleration from 0 to 30 seconds, before coming to a stop at t seconds. Calculate the total distance travelled on the journey.

[2]

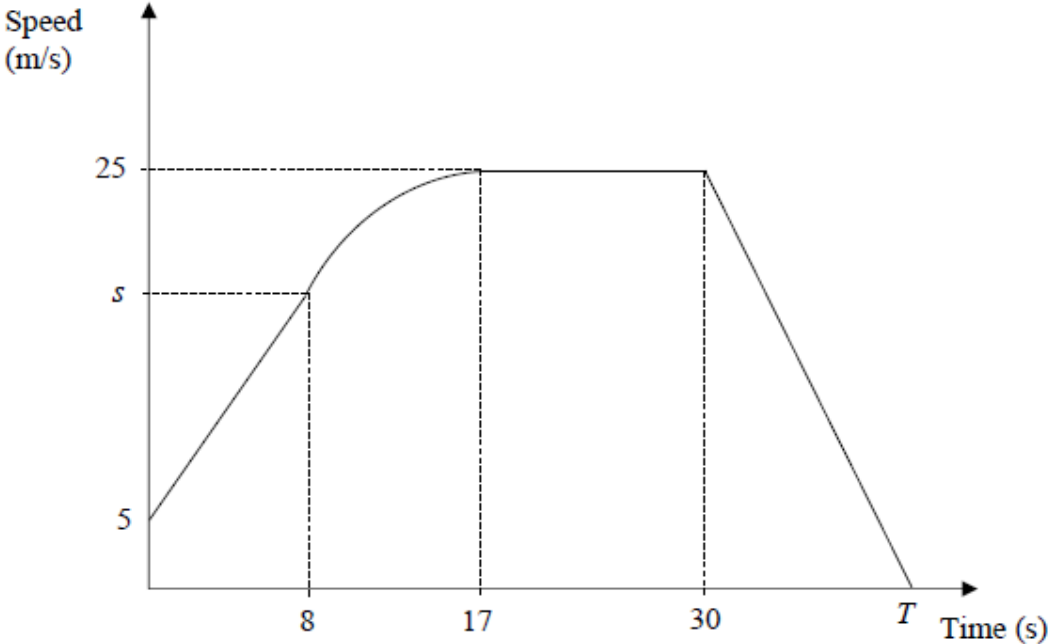
(c) Draw the distance-time graph of the object on the grid given below. You must label the values on the distance-axis clearly.



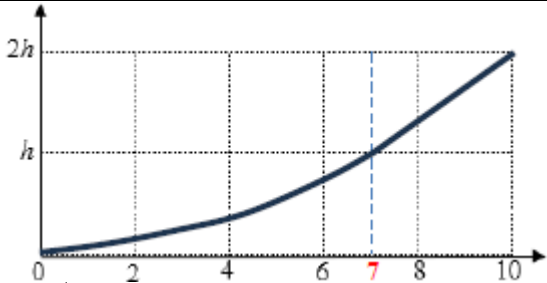
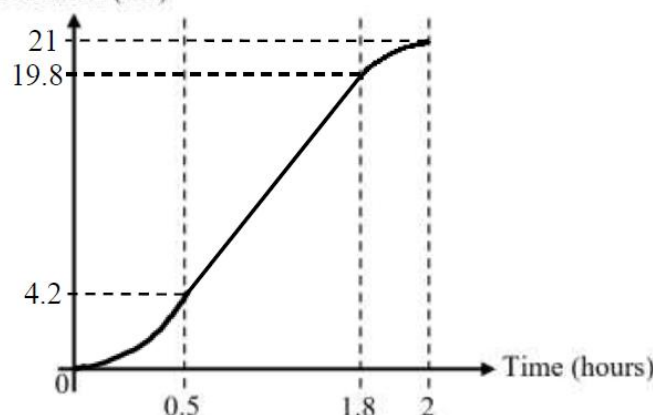
[2]

5	SST/2025/I/19
	A water tank is formed from a cylinder and a cone. The cylinder has radius r cm and height h cm. The cone has a base radius of r cm and height $6h$ cm. The water tank is filled with water to the brim of the tank. Water is drained out at a constant rate and the tank is completely emptied in 60 minutes.  (a) Find the time taken, in minutes, for the water in the cylinder to drain completely. [2] (b) On the grid provided, sketch the graph of the height of the water level in the tank against the time taken (in min). [2]
6	GMSS/2025/I/19
	The distance-time graphs of a car and a bus travelling from Town A to Town B are given below.  (a) From 0830 to 0900, which vehicle was travelling at a higher speed? Explain why. [2] (b) Calculate the average speed of the bus. [2] (c) What can you say about the average speed of the car and the bus? Explain your answer. [2]

7	GMSS/2025/I/20 It is given that $\text{US\\$}1 = \text{S\\$}1.75$. (a) Use the grid below to draw the conversion graph between US dollars and Singapore dollars. (b) Find the value of $\text{S\\$}21$ in $\text{US\\$}$.
	[1] [1]
8	ACSB/2025/I/04 Alex travels from Singapore to New Zealand. He wants to change 790 Singapore dollars (\$) into New Zealand dollars (NZD). The exchange rate in Singapore is $\\$1 = 1.29 \text{ NZD}$. He will receive 20 NZD less if he changes the money in New Zealand. Calculate the exchange rate in New Zealand. Give your answer in NZD per Singapore dollar.
	[2]

9	ACSB/2025/I/22
	The diagram below shows the speed-time graph for a car travelling on the road.  (a) The acceleration of the car in the first 8 seconds is 2m/s^2. Calculate the value of s. [1] (b) Describe the motion of the car between 8 and 17 seconds. [1] (c) The total distance travelled by the car from 17 to T seconds is 500 metres. Calculate the value of T. [2]
10	PHS/2025/II/2(C)
	A group of students travelled to Malaysia. They exchanged Singapore dollars (SGD) to Malaysian ringgit (MYR) at a rate of SGD 1 = MYR 3.20. They spent MYR 2525 on accommodation and food. They also bought 80 litres of fuel at MYR 2.50 per litre. They exchanged the remaining MYR at a rate of SGD 1 = MYR 3.25 and received SGD 60. Calculate the amount of SGD they exchanged at the start of their trip. [3]

Solution

1	(a) 0.5 (b) $0.5 \times 4 = 2$ Speed = $15 - 2 = 13 \text{ m/s}$ [B1] (c) Time taken to stop = $10 \div 0.4 = 25 \text{ s}$ Distance needed to stop = $0.5 \times 25 \times 10 = 125 \text{ m} > 100 \text{ m}$ [M1] Since the distance needed to stop is more than 100 m, he will not be able to stop in time. [A1]
2	
3	a Let the speed at $t = 0.2$ be $v \text{ km/h}$. $\frac{v - 4.8}{0.2} = \frac{12 - 4.8}{0.5}$ $v = 0.2(14.4) + 4.8$ $\therefore v = 7.68$ b $\therefore$ Average speed $= \frac{0.5(0.5)(4.8 + 12) + (12)(1.3) + 0.5(0.2)(12)}{2}$ $= \frac{4.2 + 15.6 + 1.2}{2}$ $= \frac{21}{2}$ $= 10.5 \text{ km/h}$ c 

4

(a)

$$\text{Speed} = \frac{18}{30} \times 18 = 10.8 \text{ m/s}$$

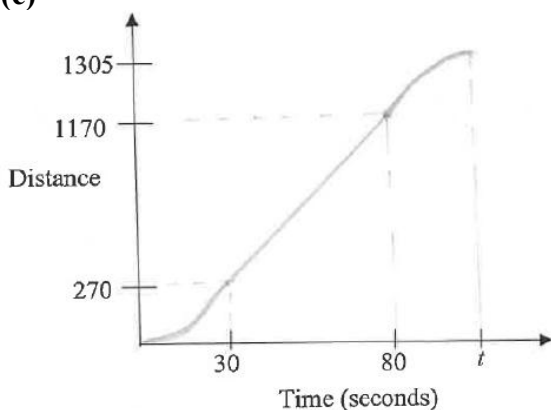
$$= 38.88 \text{ km/h}$$

(b)

$$\text{Distance} = \frac{1}{2} \times 30 \times 18 + 50 \times 18 + \frac{1}{2} \times 15 \times 18$$

$$= 1305 \text{ m}$$

(c)



5

(a)

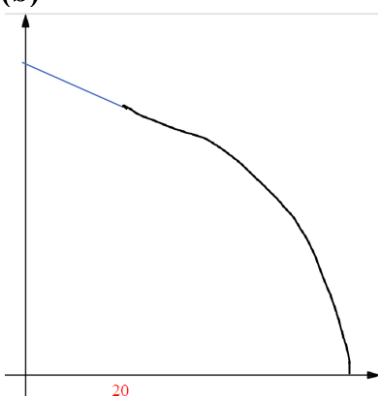
$$\text{Volume of cylinder} = \pi r^2 h$$

$$\text{Volume of cone} = \frac{1}{3} \pi r^2 6h = 2\pi r^2 h$$

Time for water in the cylinder to leak completely

$$= \frac{\pi r^2 h}{3\pi r^2 h} \times 60 = 20 \text{ minutes}$$

(b)



6

(a)

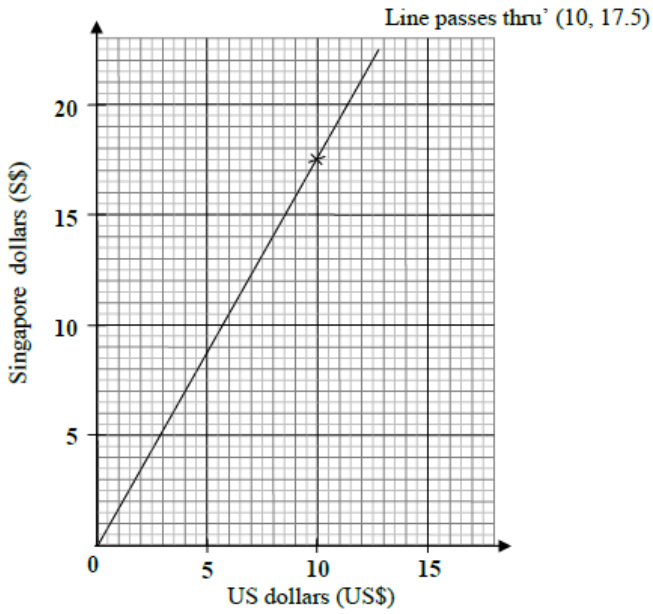
The car was travelling at a higher speed because gradient of the line for the car is steeper.

(b)

$$30\text{km} \div 2 \text{ hours} = 15 \text{ km/h}$$

(c)

The average speed of the car is higher because it takes a shorter time to complete the same journey.

7	(a)  (b) US\$12
8	$\frac{(790 \times 1.29) - 20 = 999.10}{790} = 1.26$
9	(a) $\frac{s-5}{8} = 2$ $s = 21$ (b) Car is accelerating at a decreasing rate. Speed is increasing at a decreasing rate. Rate of increase in speed is decreasing. (c) $25(30-17) + \frac{1}{2}(25)(T-30) = 500$ $T = 44$
10	Fuel cost $= 80 \times 2.50$ $= 200 \text{ MYR}$ Total Spending $= 2525 + 200 + (3.25 \times 60)$ $= 2920 \text{ MYR}$ Initial SGD $= 2920 \div 3.20$ $= 912.50 \text{ SGD}$